

25CS1302E - DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED BACKEND DEVELOPMENT

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SECTION: 2A

LAB COURSEWORK

Q. Week 1:

A local library is currently managing its books and members using a physical notebook, which has resulted in several operational problems such as misplaced books, difficulty tracking borrowed items, and inaccurate member records. To modernize its operations, the library plans to develop a database system called **BookFlow**.

As a database developer, your task is to create the SQL foundation for this system.

First, create a new database

named **bookflow_db**. Then, design two tables: a **Books** table containing the columns **book_id**, **title**, **isbn**, and

published_year, and a **Members** table containing the columns **member_id**, **full_name**, and **email**. Ensure data

integrity by setting **book_id** and **member_id** as **PRIMARY KEYs**, applying a **NOT NULL** constraint to the

title column so that every book must have a name, enforcing **UNIQUE** constraints on both **isbn** and **email** to

prevent duplicate entries, and adding a **CHECK** constraint to ensure that the **published_year** is not in the future

(for example, it must be less than 2027). Finally, insert three sample records into both the **Books** and **Members**

tables to verify that the database structure and constraints are functioning correctly. This exercise will help you

develop skills in schema definition, data integrity enforcement, and the use of SQL Data Definition Language

(DDL) commands such as **CREATE DATABASE**, **CREATE TABLE**, and **INSERT** statements

SOL: % mysql -u root -p

Enter password:

Welcome to the MySQL monitor. Commands end with ; or \g.

Your MySQL connection id is 28

Server version: 9.7.1 Homebrew

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> SHOW DATABASES;

```
+-----+
| Database      |
+-----+
| college       |
| information_schema |
| mysql         |
| performance_schema |
| sys           |
| test          |
+-----+
```

6 rows in set (0.030 sec)

mysql> CREATE DATABASE IF NOT EXISTS bookflow_db;

Query OK, 1 row affected (0.018 sec)

mysql> USE bookflow_db;

Database changed

mysql> SELECT DATABASE();

```
+-----+
| DATABASE() |
+-----+
| bookflow_db |
+-----+
```

1 row in set (0.000 sec)

mysql> CREATE TABLE books (

```
->  book_id INT AUTO_INCREMENT PRIMARY KEY,
->  title VARCHAR(255) NOT NULL,
->  isbn VARCHAR(13) NOT NULL UNIQUE,
->  published_year INT,
->  CONSTRAINT chk_published_year CHECK (published_year < 2027)
-> );
```

Query OK, 0 rows affected (0.022 sec)

```
mysql> SHOW TABLES;
```

```
+-----+  
| Tables_in_bookflow_db |  
+-----+  
| books                  |  
+-----+
```

```
1 row in set (0.002 sec)
```

```
mysql> desc books;
```

```
+-----+-----+-----+-----+-----+-----+  
| Field      | Type      | Null | Key | Default | Extra      |  
+-----+-----+-----+-----+-----+-----+  
| book_id    | int       | NO   | PRI | NULL    | auto_increment |  
| title      | varchar(255) | NO   |     | NULL    |               |  
| isbn       | varchar(13) | NO   | UNI | NULL    |               |  
| published_year | int       | YES  |     | NULL    |               |  
+-----+-----+-----+-----+-----+-----+
```

```
4 rows in set (0.007 sec)
```

```
mysql> CREATE TABLE members (
```

```
-> member_id INT AUTO_INCREMENT PRIMARY KEY,  
-> full_name VARCHAR(100) NOT NULL,  
-> email VARCHAR(150) NOT NULL UNIQUE  
-> );
```

```
Query OK, 0 rows affected (0.010 sec)
```

```
mysql> DESCRIBE members;
```

```
+-----+-----+-----+-----+-----+-----+  
| Field      | Type      | Null | Key | Default | Extra      |  
+-----+-----+-----+-----+-----+-----+  
| member_id | int       | NO   | PRI | NULL    | auto_increment |  
| full_name | varchar(100) | NO   |     | NULL    |               |  
| email     | varchar(150) | NO   | UNI | NULL    |               |  
+-----+-----+-----+-----+-----+-----+
```

```
3 rows in set (0.003 sec)
```

```
mysql> INSERT INTO books (title, isbn, published_year)
```

```
-> VALUES  
-> ('The Alchemist', '9780061122415', 1988),  
-> ('Clean Code', '9780132350884', 2008),  
-> ('Atomic Habits', '9780735211292', 2018);
```

```
Query OK, 3 rows affected (0.005 sec)
```

```
Records: 3 Duplicates: 0 Warnings: 0
```

```
mysql> select * from books;
```

book_id	title	isbn	published_year
1	The Alchemist	9780061122415	1988
2	Clean Code	9780132350884	2008
3	Atomic Habits	9780735211292	2018

3 rows in set (0.001 sec)

```
mysql> INSERT INTO members (full_name, email)
-> VALUES
-> ('Anil Kumar', 'anil.kumar@example.com'),
-> ('Priya Sharma', 'priya.sharma@example.com'),
-> ('Ravi Verma', 'ravi.verma@example.com');
```

Query OK, 3 rows affected (0.003 sec)

Records: 3 Duplicates: 0 Warnings: 0

```
mysql> select * from members;
```

member_id	full_name	email
1	Anil Kumar	anil.kumar@example.com
2	Priya Sharma	priya.sharma@example.com
3	Ravi Verma	ravi.verma@example.com

3 rows in set (0.000 sec)

```
mysql> INSERT INTO books (title, isbn, published_year)
```

```
-> VALUES ('Fake Copy', '9780061122415', 2000);
```

ERROR 1062 (23000): Duplicate entry '9780061122415' for key 'books.isbn'

```
mysql> INSERT INTO books (title, isbn, published_year)
```

```
-> VALUES (NULL, '9999999999999', 2010);
```

ERROR 1048 (23000): Column 'title' cannot be null

```
mysql> INSERT INTO books (title, isbn, published_year)
```

```
-> VALUES ('Time Traveler', '8888888888888', 2030);
```

ERROR 3819 (HY000): Check constraint 'chk_published_year' is violated.

```
mysql> INSERT INTO members (full_name, email)
```

```
-> VALUES ('Anil Clone', 'anil.kumar@example.com');
```

ERROR 1062 (23000): Duplicate entry 'anil.kumar@example.com' for key 'members.email'

```
mysql>
```